What makes it an autonomous digital twin is that it’s not just logging data or reacting blindly. It’s assimilating sensor data into its model and acting autonomously to improve comfort and save energy. You’re actually living with a low-key autonomous digital twin right there in your home.

1.4 What are digital twins good for?

Hopefully, you are starting to form a picture in your mind as to how you might currently benefit from the digital twins we have touched on so far, but as someone looking to build and deploy a digital twin within their business, what are the potential benefits that they offer? Let’s explore some of these key advantages.

1.4.1 Accelerating product development

This pre-construction model allows engineers to test and simulate countless outcomes and "what-if" scenarios in a risk-free environment. By running simulations, developers can quickly identify performance limitations, predict specific failure modes, and refine their designs. This approach alters the traditional product development cycle, moving away from expensive, iterative physical prototyping. Instead, it enables the rapid testing of many design variants at a far lower cost, allowing physical tests to be focused only on the most promising designs and predicted critical failure points, thereby accelerating innovation.

REAL WORLD EXAMPLE: SIEMENS GAS TURBINE DEVELOPMENT

When Siemens developed the SGT-A65 gas turbine, they adopted a digital twin known as ATOM (Agent-based Turbine Operations and Maintenance). ATOM modeled the fleet operations of Siemens gas turbines, including engine characteristics, supply chain logistics, maintenance facility operations, and customer operations. Replacing a set of Excel-based forecasting tools and representing the entire system, ATOM reportedly provided enhanced analytical capabilities and enabled decision making that considered the entire system of turbine operations.